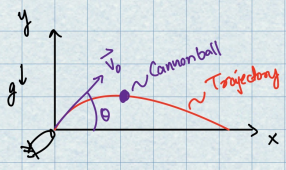


# 1 Ballistics

## 1.1 Problem and Dynamics:

# Ballistics w/ Linear Drag



FBD of cannonball

$$\vec{F}_d = c \vec{v} \hat{v} = c \vec{v} \frac{-\vec{v}}{v} = -c \vec{v}, \text{ where } \vec{v} \text{ is velocity vector}$$

Apply LMB,  $\sum \vec{F} = m \vec{a}$

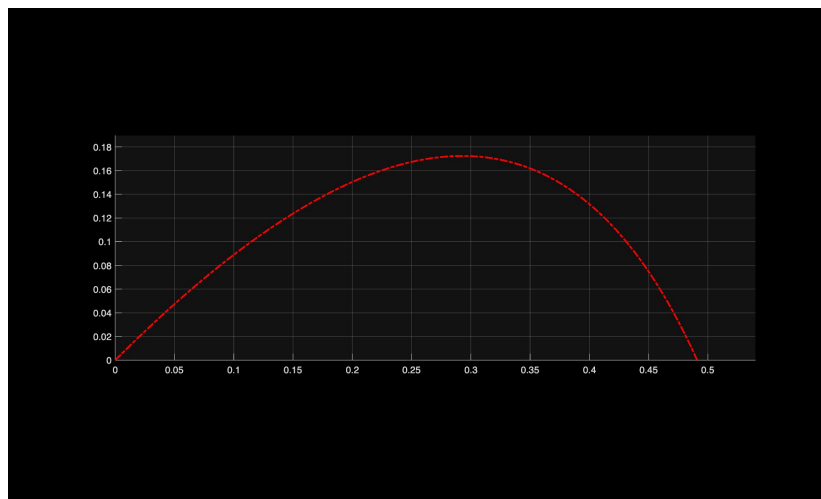
EoM:  $\frac{-1}{m} (mg \hat{j} + c \vec{v}) = \vec{a}$

States:  $[\vec{x}; \vec{v}] = [x; y; v_x; v_y]$

initial condition =  $[x; y; v_0 \cos \theta; v_0 \sin \theta]$  for given,  $\vec{x}, v_0, \theta$

Dynamics Derivation

## 1.2 All params set to 1 and theta = 45 deg



### 1.3 Increasing $V_0$

